

Sardar Vallabhbhai National Institute of Technology (SVNIT), Surat
Department of Computer Science and Engineering
B.Tech II - Semester-III
CS231 – Object Oriented Programming Practical Assignment
Lab Assignment - 12

(1). Write a C++ program to create a simple system for calculating the area of different shapes (Circle and Rectangle) that demonstrates:

- Pointer to Function – to call functions like inputData() or displayResult().
- Pointer to Data Member – to access or modify the dimensions of the shapes.
- Pointer to Object – to access members of objects dynamically.
- this Pointer – to display the current object's details.
- Virtual Function – to define a common area() function in the base class.
- Runtime Polymorphism – to call the derived class area() functions through a base class pointer.

(2) Write a C++ program to manage employee details using classes and dynamic memory allocation. Create a class Employee with data members name, age, and salary. Include functions to: Input and display employee details (use the this pointer to show the current object); and update salary by a given percentage. In main(), dynamically create employee objects using pointers, call member functions through them, and demonstrate the use of the this pointer.

(3) Write a C++ program to calculate vehicle service cost using virtual functions. Create a base class Vehicle with a virtual function serviceCost() and a normal function displayType(). Derive two classes: *Car* – $cost = baseCost + (engineCC * 0.5)$ and *Bike* – $cost = baseCost + (engineCC * 0.2)$. In main(), use base class pointers to call the functions for Car and Bike objects and demonstrate runtime polymorphism.

(4) Write a C++ program to develop a simple Banking System - **that demonstrates:**

- **Pointer to Function** – to call operations such as **deposit()**, **withdraw()**, and **displayBalance()**.
- **Pointer to Data Member** – to update the balance of an account.
- **Pointer to Object** – to create and access multiple **Account** objects dynamically.
- **this Pointer** – to identify the current active account.
- **Virtual Function and Runtime Polymorphism** – to implement a base class **Account** with a virtual **calculateInterest()** function, overridden in derived classes **SavingsAccount** and **CurrentAccount**.

Requirements:

1. Create a base class **Account** with members: **name**, **accNo**, and **balance**.
2. Create derived classes **SavingsAccount** and **CurrentAccount**.
3. Use runtime polymorphism to call the **calculateInterest()** function for each type of account through a base class pointer.
4. Demonstrate all pointer types as listed above.

